Flashcard Quizzer: Consolidated Scan and Test Report

Repository	github.com/dipanjan-acn/ud-ainse-assignment
Commit reviewed	2e8b2ff (main), after the security fixes
Platform	macOS 26.7.1, Python 3.11.17
Date	2026-10-24

> **Overall: PASS**
> Every scan that ran passes on the current commit. Three scan findings were fixed: bandit (first round), pip-audit (first round), and bandit B310 (latest round). The safety scan could not run without an account, so it was removed from the requirements.

1. Consolidated scan results

"Before" is the first run in this review. "After" is the run on the current commit. Actions are the changes made as a result.

Area	Tool	Before	After	Status	Action taken
Formatting	black --check	Clean	Clean	PASS	None needed
Import order	isort --check-only	Clean	Clean	PASS	None needed
Lint	flake8 (88-column limit)	0 issues	0 issues	PASS	None needed
Types	mypy --strict	0 errors	0 errors	PASS	None needed
Tests and coverage	pytest, branch coverage 100% enforced	204 passed	204 of 204 passed	PASS	None needed
Security (static)	bandit, app code (867 LOC)	2 low findings: B311 in cli.py and quiz_engine.py	0 findings	PASS	Shuffle generator moved to new_generator() with a # nosec B311 justification
Security (static), latest round	bandit after the AI network code was added	1 medium: B310, urlopen with a caller-supplied URL	0 findings	PASS	Added an HTTPS-only check before urlopen, with tests. The nosec comment is kept only after the check
Dependencies	pip-audit against requirements.txt	3 advisories in filelock 3.32.7 (fix: 4.2.1), 87 packages	0 advisories, 76 packages	PASS	Pinned filelock>=4.2.1; added pip-audit as the scanner
Dependencies (account)	safety scan	Not run: 'check' deprecated; 'scan' needs a Safety account login	Not run	NOT RUN	Removed from requirements.txt: needs login, and it conflicts with filelock 4.x
Scripted manual checks	scripts/manual_tty_checks.py	Earlier run: 14 of 14 (before config and generation scenarios)	17 of 17 passed	PASS	None needed

2. Actions taken

#	Finding	Action	Verified by
1	bandit B311 (low): standard random generator used on two lines	Moved to a single `new_generator()` helper, with a `# nosec B311` comment and a note that card order is not security-sensitive	bandit: 0 issues on 535 LOC
2	pip-audit: three advisories in filelock 3.32.7 (CVE-2026-112907, CVE-2026-112913, CVE-2026-112940)	Pinned `filelock>=4.2.1`. The package is a transitive development dependency, not used by the app at runtime	pip-audit: 0 advisories; fresh install from requirements.txt succeeds

3	safety: the 'check' command is deprecated, and 'scan' requires a Safety account login	Removed from requirements.txt. pip-audit replaces it, because it needs no account. Safety also conflicted with filelock 4.x	Fresh install resolves with no conflicts
4	bandit B310 (medium): urlopen could open a file or other scheme if the URL was not fixed	Added an HTTPS-only check before urlopen. Tests confirm http, file, and ftp URLs are refused before any request	bandit: 0 issues on 867 LOC; tests for each refused scheme
5	Requirements file error found during the dry-run: the filelock line was joined onto the pydantic line, which turned pydantic into a comment	Split the lines and restored the pydantic requirement	Fresh install from requirements.txt; pip-audit run afterwards

3. Quality gates

Gate	Result
black --check	PASS
isort --check-only	PASS
flake8	PASS
mypy strict	PASS
coverage 100 percent	PASS

Coverage: `TOTAL 487 0 88 0 100%`

4. Automated tests by file

Test file	Passed	Failed	Result
test_ai_generator	30	0	PASS
test_cli	16	0	PASS
test_cli_settings	9	0	PASS
test_code_limits	31	0	PASS
test_config	17	0	PASS
test_display	14	0	PASS
test_export	5	0	PASS
test_flashcard_loader	26	0	PASS
test_input_limits	7	0	PASS
test_integration	4	0	PASS
test_observers	7	0	PASS
test_quiz_engine	15	0	PASS
test_quiz_modes	15	0	PASS
test_quiz_runner	8	0	PASS
Total	**204**	**0**	PASS

5. Manual scenarios

Each scenario runs the app and checks its exit code and output. Terminal scenarios run in a pseudo-terminal, so colors and Ctrl+C behave as they do for a user.

PASS **TTY-01: Sequential session, one miss, colors on**

Command: `python main.py --mode sequential --file data/glossary.json --stats`
Expected: Exit 0; 9 green and 1 red feedback line; summary of 10 attempts at 90.0%

- ✔ PASS: exit code is 0
- ✔ PASS: 9 green feedback lines
- ✔ PASS: 1 red feedback line
- ✔ PASS: summary shows 10 attempts
- ✔ PASS: accuracy is 90.0%

PASS **TTY-02: NO_COLOR environment variable disables colors**

Command: `python main.py -f data/glossary.json`
Expected: Exit 0; no escape codes in the output

- ✔ PASS: exit code is 0
- ✔ PASS: no escape codes

PASS **TTY-03: --no-color flag disables colors**

Command: `python main.py -f data/glossary.json --no-color`
Expected: Exit 0; no escape codes in the output

- ✔ PASS: exit code is 0
- ✔ PASS: no escape codes

PASS **TTY-04: Adaptive session repeats a missed card**

Command: `python main.py -m adaptive -f data/python_basics.json --stats --no-color`
Expected: Exit 0; 9 attempts for 8 cards, 8 correct

- ✔ PASS: exit code is 0
- ✔ PASS: 9 attempts
- ✔ PASS: 8 correct

PASS **TTY-05: Typing 'exit' stops the quiz early**

Command: `python main.py -f data/glossary.json --no-color`
Expected: Exit 0; 'Leaving the quiz early.' is printed

- ✔ PASS: exit code is 0
- ✔ PASS: early-exit message

PASS **TTY-06: Real Ctrl+C stops gracefully**

Command: `python main.py -f data/glossary.json`
Expected: Exit 0; 'Quiz stopped. Goodbye!' printed; no traceback

- ✔ PASS: exit code is 0
- ✔ PASS: goodbye message
- ✔ PASS: no traceback

PASS **CLI-01: --help lists every flag**

Command: `python main.py --help`
Expected: Exit 0; help text names all six flags

- ✔ PASS: exit code is 0
- ✔ PASS: lists --file
- ✔ PASS: lists --mode
- ✔ PASS: lists --stats
- ✔ PASS: lists --export
- ✔ PASS: lists --log-file
- ✔ PASS: lists --no-color

PASS **CLI-02: Card without 'back' gives a one-line error**

Command: `python main.py -f /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/missing_back.json`
Expected: Exit 1; 'Error:' names the field; no traceback

- ✔ PASS: exit code is 1
- ✔ PASS: names 'back'
- ✔ PASS: no traceback

PASS **CLI-03: Malformed JSON gives a friendly error**

Command: `python main.py -f /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/broken.json`
Expected: Exit 1; 'Invalid JSON' with a line number

- ✔ PASS: exit code is 1
- ✔ PASS: invalid JSON message
- ✔ PASS: no traceback

PASS **CLI-04: Missing deck file gives a friendly error**

Command: `python main.py -f /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/no_such_deck.json`
Expected: Exit 1; 'File not found' message

- ✔ PASS: exit code is 1
- ✔ PASS: file-not-found message

PASS **CLI-05: Export to a folder fails before the quiz starts**

Command: `python main.py -f data/glossary.json --export /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35`
Expected: Exit 1; error printed; no question is asked

- ✔ PASS: exit code is 1
- ✔ PASS: error message
- ✔ PASS: no question asked

PASS **CLI-06: Adaptive session, all cards correct (piped input)**

Command: `python main.py -m adaptive -f data/python_basics.json --no-color --stats`
Expected: Exit 0; 100.0% accuracy

- ✔ PASS: exit code is 0
- ✔ PASS: accuracy is 100.0%

PASS **CLI-07: Export and log file written during a session**

Command: `python main.py -m adaptive -f data/python_basics.json --no-color --export /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/report.json --log-file /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/answers.log`
Expected: Exit 0; report has 2 attempts; log has 2 lines

- ✔ PASS: exit code is 0
- ✔ PASS: report has 2 attempts
- ✔ PASS: log has 2 lines

PASS **CLI-08: Success-criteria command: sequential mode, glossary deck**

Command: `python main.py --mode sequential --file data/glossary.json`
Expected: Exit 0; quiz starts and exits cleanly

- ✔ PASS: exit code is 0
- ✔ PASS: first card shown

PASS **CLI-09: Invalid mode in FLASHCARD_MODE gives a clear error**

Command: `python main.py -f data/glossary.json`
Expected: Exit 1; names the unknown mode

- ✔ PASS: exit code is 1
- ✔ PASS: names the mode

PASS **CLI-10: --generate without an API key stops before any network call**

Command: `python main.py --generate networking --save /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/generated.json`
Expected: Exit 1; message names ANTHROPIC_API_KEY; no deck written

- ✔ PASS: exit code is 1
- ✔ PASS: names the variable
- ✔ PASS: no deck written

PASS **CLI-11: --generate without --save is refused**

Command: `python main.py --generate networking`
Expected: Exit 1; asks for --save

- ✔ PASS: exit code is 1
- ✔ PASS: asks for --save

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6. Limitations

- Windows was not tested. Terminal scenarios would be reported as SKIPPED there, and console colors have not been checked.
- The safety scan did not run. Its replacement, pip-audit, uses the same PyPI advisory data but may differ from Safety's own database.
- Only the app code was scanned by bandit. The tests and the manual script were excluded.
- Scans cover this commit only. Earlier commits were not re-scanned.